CleanGS: Pareto-Guided Post-Hoc Cleanup of 3D Gaussian Splatting Scenes

Supplementary Material

Jyotirmaya Shivottam
National Institute of Science Education and Research
Bhubaneswar, Odisha, India
Homi Bhabha National Institute
Mumbai, India
jyotirmaya.shivottam@niser.ac.in

Om Kumar
Manipal Institute of Technology
Manipal, Karnataka, India
om.mitmpl2023@learner.manipal.edu

Subhankar Mishra
National Institute of Science Education and Research
Bhubaneswar, Odisha, India
Homi Bhabha National Institute
Mumbai, India
smishra@niser.ac.in

In this supplement, *standard* (or *captured*, 1×) refers to evaluation at the original camera framing, and *widened* (or *zoomed*, 6×) to the fixed-pose evaluation with the focal length divided by six. In tables, the subscript 6× and the label "Zoom" refer to the widened setting.

1 PROOFS

1.1 Dominance is preference-free

We restate the dominance lemma, Lemma 3.1 (Main paper), here. Throughout this analysis the feasible set consists of the k-subsets of the reduced candidate pool $\mathcal{V}_r$ that the selector actually enumerates. Let $f(S) = (A(S), B(S), O(S), T(S))$ and let S_2 *dominate* S_1 when $f(S_2) \geq f(S_1)$ coordinatewise with at least one strict inequality. Let $\mathcal{F}$ be the set of non-dominated subsets.

Lemma 1.1. *Let $U(S) = \Phi(f(S))$ for any $\Phi\colon \mathbb{R}^4 \to \mathbb{R}$ that is non-decreasing in each argument. If S_2 dominates S_1 then $U(S_2) \geq U(S_1)$, and $\max_S U(S)$ is attained on $\mathcal{F}$.*

Proof. Write $f(S_1) = (x_1, \ldots, x_4)$ and $f(S_2) = (y_1, \ldots, y_4)$ with $y_\ell \geq x_\ell$ for every ℓ. Replace the coordinates one at a time. Monotonicity of Φ in its ℓ-th argument gives

$$\Phi(y_1, \ldots, y_{\ell-1}, x_\ell, \ldots, x_4) \leq \Phi(y_1, \ldots, y_{\ell-1}, y_\ell, x_{\ell+1}, \ldots, x_4)$$

for each ℓ, and chaining the four inequalities yields $\Phi(f(S_1)) \leq \Phi(f(S_2))$.

For the second claim, let S be any subset. If $S \in \mathcal{F}$ there is nothing to prove. Otherwise S is dominated, and since the number of feasible subsets is finite, following a chain of dominating subsets terminates at some $S' \in \mathcal{F}$. Each step is non-decreasing in U, so $U(S') \geq U(S)$. Hence no subset outside $\mathcal{F}$ attains a value larger than the maximum over $\mathcal{F}$. □

The weighted Chebyshev rule used in the main paper is one such Φ after sign reversal, being non-increasing in each normalized shortfall $z_\ell - \widetilde{f}_\ell(S)$ and hence non-decreasing in each objective. The weights therefore express a preference among mutually incomparable subsets, and they cannot select a subset that another subset beats on all four objectives simultaneously.

1.2 Independent support suppresses clutter

The support proposition, Proposition 3.2 (Main paper), is restated and proved here.

Proposition 1.2. *Let*

$$p_f = \Pr[X_v(g) = 1 \mid g \text{ foreground}],$$
$$p_b = \Pr[X_v(g) = 1 \mid g \text{ background}],$$

with $p_f > p_b$, and assume the k support indicators $\{X_v(g)\}_{v \in S^\star}$ are conditionally independent and identically distributed given the class of g. Under the keep rule $\nu(g) \geq \tau$,

$$\Pr[\nu(g) \geq \tau \mid g] = \sum_{j=\tau}^{k} \binom{k}{j} p^j (1-p)^{k-j},$$

with $p = p_f$ for foreground and $p = p_b$ for background. For $k = 3$ and $\tau = 2$,

$$\Pr[\nu(g) \geq 2 \mid g \text{ background}] = 3p_b^2(1-p_b) + p_b^3 \leq 3p_b^2.$$

Proof. Under the stated assumptions $\nu(g)$ is a sum of k conditionally i.i.d. Bernoulli indicators, hence binomial with parameters (k, p_f) for foreground and (k, p_b) for background. The retention probabilities are the corresponding upper tails, which gives the displayed sum.

For the bound, set $k = 3$ and $\tau = 2$ and expand:

$$\sum_{j=2}^{3} \binom{3}{j} p_b^j (1-p_b)^{3-j} = 3p_b^2(1-p_b) + p_b^3 = 3p_b^2 - 2p_b^3.$$

Since $p_b \geq 0$, $2p_b^3 \geq 0$, and therefore $3p_b^2 - 2p_b^3 \leq 3p_b^2$ for all $p_b \in [0, 1]$. □

Dependence between views. Proposition 1.2 is an idealized reference model rather than an assumption that real capture views can satisfy exactly. Two views with similar camera centers, viewing directions, and masked support can produce strongly correlated support indicators. For example, if $X_1 = X_2$ and X_3 is independent, a two-of-three rule fires whenever the duplicated observation is positive, with probability p, rather than the p^2 predicted by a two-trial binomial at the same threshold. The diversity objectives B,

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
ICVGIP2026, December 21–24, 2026, Kolkata, India
© 2026 Copyright held by the owner/author(s).
ACM ISBN 978-x-xxxx-xxxx-x/YY/MM.
https://doi.org/10.1145/nnnnnnn.nnnnnnn

Table 1: Empirical dependence of per-view support. Support indicators $X_v(g)$ of the three filtering views over 18 scenes. Random subsets use seeds 0–2, averaged within scene; two *table* runs with an unregistered camera are excluded. p is a two-sided Wilcoxon signed-rank test.

Measure	Selected	Random	p
Pairwise ϕ, co-visible Gaussians (mean)	0.61	0.51	0.32
$\Pr[\nu \geq 2]$ / independence prediction (mean)	3.3	2.6	0.09
Supported by exactly one view (fraction)	0.27	0.45	0.001

T, and O are therefore intended to discourage repeated evidence, while A favors views whose semantic support is reliable.

Measured dependence. To measure this dependence, we recorded the per-view support indicator $X_v(g)$ of every Gaussian during filtering; recording does not change the output. Table 1 compares the selected views with random three-view subsets (seeds 0–2, averaged within scene). Support is strongly positively dependent: over co-visible Gaussians, the mean pairwise ϕ coefficient is 0.61, and two-of-three agreement occurs at a median of 1.99× the rate predicted under independence. Selected views also share more support than random views: 27% of supported Gaussians are supported by exactly one selected view, against 45% for random views ($p = 0.001$, lower on 15 of 18 scenes). This shared support is consistent with the higher retention and target fidelity of selected views (Section 9). Proposition 1.2 should therefore be read as an idealized motivation for the vote rather than as a quantitative prediction.

The vote threshold also matters when k changes. At a fixed absolute threshold, $\Pr[\mathrm{Bin}(3,p) \geq 2] = 3p^2 - 2p^3 > p^2 = \Pr[\mathrm{Bin}(2,p) \geq 2]$, so simply adding votes can increase retention of both foreground and background. This is the behavior seen in the budget sweep for a fixed two-of-k rule and motivates considering view budget and threshold together.

Regime captured by the bound. The quadratic term is most informative when p_b is small, as for free-space floaters or clutter whose projection falls inside the target mask from only a limited set of directions. Target-adjacent structure is a harder regime: when background remains aligned with the target across views, p_b approaches p_f and semantic support alone provides less class separation. This explains the residual structures highlighted in the main paper and motivates complementary geometric cues. The clustering study in Section 13 shows that spatial connectivity can substantially reduce leakage, while a global largest-component rule becomes unreliable when the retained target is distributed across multiple density modes. Renderer visibility introduces a related ambiguity, because it identifies potential contribution to a render rather than target-side geometric occlusion.

2 VIEW-SELECTION DETAILS

2.1 Per-view band-pass scoring

Each candidate view v is scored by

$$a_v = \exp\left(-\tfrac{1}{2}\left(\tfrac{r_v - r_0}{\sigma}\right)^2\right), \qquad r_v = \frac{|M_v|}{|\Omega_v|},$$

with $r_0 = 0.30$ and $\sigma = 0.15$. The score peaks at intermediate occupancy rather than increasing with mask area, because occupancy is not monotone with usefulness for semantic lifting.

At very small r_v, the target occupies few pixels, so a fixed image-space boundary error represents a larger fraction of the silhouette. These views also contain fewer masked sparse tracks, making Q_v a weaker summary of the target's reconstructed support. At very large r_v, the candidate is often a close view with limited context or target truncation at the image boundary. Intermediate occupancy therefore provides a useful operating region for semantic lifting.

The hard containment tests complement the smooth occupancy score by detecting truncation directly. Masks are rejected when they touch more than one image border, approach the frame too closely, or nearly fill either image dimension. Together, the soft band-pass score and hard containment tests favor views that show a substantial, well-contained target with enough surrounding context for multiview discrimination.

Concretely, we retain candidates whose raw occupancy lies in $[0.02, 0.60]$, allow at most one touched image border, require a normalized bounding-box margin of at least 0.03, and reject masks whose normalized bounding box exceeds 0.90 in either image dimension. We then apply the mask-level dominance filter over reliability, border margin, and truncation, giving the reduced pool $\mathcal{V}_r$ used for subset enumeration. Two fallbacks keep this deterministic. If the dominance filter would leave fewer than k candidates, it is skipped and every eligible mask is retained; at $k = 3$ this happens on *century*, *red brick build*, and *picnic*. If the selector still produces no subset, the containment tests are disabled and the run repeats with the candidate pool capped at 60 views. The cap sorts candidates by decreasing band-pass score a_v, keeps the first 60, and breaks ties by the order in which views appear in the COLMAP model, so it depends only on the scores and that ordering; at $k = 3$ this fallback runs on *century*, *red brick build*, and *table*, and the cap removes views only on *century* and *red brick build*. The primary path enumerates the reduced pool uncapped.

Candidate counts. Across the 18 scenes, the selector scores a median of 301 candidate masks (range 268–358). A median of 72.5 remain after the containment tests (0–258) and 10.5 after the dominance filter (3–268), so the enumerated pool has a median of 10.5 views (3–60). This gives a median of 142.5 scored triples (1–34,220) and a Pareto front of 13 subsets (1–472). The Pareto front in the main paper is therefore exact over this reduced pool.

2.2 The four subset objectives

Coverage $A(S)$. The minimum over the subset provides a weakest-link reliability criterion. A heavily truncated or under-segmented view can contribute systematically incomplete evidence, while a mean can hide it behind two strong views. Using the minimum requires every selected view to clear the same basic semantic bar before it influences 3D filtering.

Baseline diversity $B(S)$. Multiview localization is governed by parallax, which vanishes when cameras are nearly co-located. Two views may both be semantically clean and visually distinct and still provide nearly identical 3D evidence if their centers are close. In that regime small image-space or mask errors produce large uncertainty

in the lifted support. Larger baselines make the corresponding rays intersect more stably in 3D [2].

Viewing-angle diversity $T(S)$. Baseline alone does not guarantee directional diversity. Cameras can move appreciably while still viewing the target from nearly the same direction, especially under forward motion or along shallow arcs, leaving the subset sampling a narrow portion of the target's appearance and surface support. Angular diversity reduces the chance that the selected views all miss the same regions or repeat the same occlusion pattern [8].

Masked-support complementarity $O(S)$. Structure-from-motion and multi-view stereo usually favor large track overlap because shared observations support matching and joint optimization [7]. Here calibration has already been established and all masks refer to the same target, so subset selection can instead reward complementary support. The Jaccard dissimilarity $O(S)$ favors pairs whose masked track sets extend one another, reducing the chance that multiple votes repeat essentially the same reconstructed evidence.

Normalization and scalarization. Each objective is min–max normalized across *all* enumerated k-subsets of $\mathcal{V}_r$, so the normalized values do not depend on which subsets turn out to be non-dominated. The weighted Chebyshev criterion is then evaluated on the normalized objectives of the front members. When an objective is constant across all subsets, its normalized column is set to one. Thus, it contributes no shortfall to the Chebyshev criterion and the remaining objectives determine the selection.

2.3 Candidate reduction, the pool cap, and direct scalarization

Direct scalarization. Lemma 1.1 implies that the Chebyshev optimum lies on the Pareto front, so the front need not be constructed to obtain $S^\star$: minimizing the criterion directly over all enumerated k-subsets returns the same subset. In practice, a direct argmin returns the selected subset on 16 of 18 scenes. The remaining two, *gardens* and *tractor*, have exact ties in the Chebyshev value (3 and 5 tied subsets), where the first tied index can be a dominated subset. Removing dominated subsets within the tie set before taking the first index recovers the selection on all 18 scenes, and on 15 further scenes outside the benchmark. Direct minimization takes under 0.1 ms at $k = 3$, whereas the all-pairs front costs about 51 s on the two scenes whose pool reaches the cap of 60 views and under 0.1 s elsewhere. We keep the explicit front because it makes the trade-offs among the objectives visible and supports the reweighting analysis, but the pipeline does not need it. The $k \geq 4$ arms in Section 11 use the direct argmin.

Pool cap and dominance filter. Only *century* (178 eligible candidates) and *red brick build* (268) have more than 60 eligible candidates, so the cap affects only these two scenes. Table 2 removes or relaxes the cap on these two scenes using the direct argmin. The effect is scene dependent and has no consistent sign. Disabling the mask-level dominance filter on the six scenes where it shrinks the pool most (e.g. *Horse*, 169 to 11 views; *tractor*, 258 to 19) changes standard PSNR by −0.40 to +0.30 dB and widened PSNR by −0.22 to +2.36 dB, and lowers leakage on all six scenes (−0.02 to −0.14). The dominance filter therefore mainly keeps the explicit front tractable; the direct argmin does not need it.

Table 2: Effect of the 60-view pool cap. Standard (captured-framing) PSNR and leakage on the two scenes with more than 60 eligible candidates; Δ is relative to the main cap of 60.

Scene	Cap 60 (main)	Cap 120	Uncapped
century	19.94 dB, 0.391	18.98 (−0.96), 0.351	18.57 (−1.37), 0.347
red brick build	16.34 dB, 0.196	16.82 (+0.48), 0.201	17.63 (+1.29), 0.207

2.4 Relation to active view selection and mask confidence

Active view-selection methods for radiance fields ask which new observation would most improve a reconstruction. ActiveNeRF selects additional views using estimated uncertainty [6], while FisherRF uses Fisher information and expected information gain for active view selection and mapping [3]. CleanGS addresses a complementary problem: the scene is already reconstructed and the candidate cameras already exist, so the selector chooses which available masked views should provide semantic evidence for post-hoc cleanup rather than which image should be acquired next. These methods nevertheless suggest uncertainty- or information-based alternatives to the geometric diversity terms studied here.

Mask confidence provides another natural alternative to occupancy-based reliability. The original Segment Anything model predicts an estimated IoU score for each candidate mask [4]; such model-derived confidence could be used directly or combined with a_v. The occupancy score used here is intentionally model-agnostic and depends only on the final binary mask and camera frame. Comparing learned mask confidence with this simple geometric proxy is a useful direction for a dedicated view-quality study.

3 COMPLEXITY

Let n_{all} be the number of capture views, n the candidates surviving reduction, $k = 3$ the subset size, $\bar{q}$ the mean number of masked sparse tracks per view, N_G the Gaussian count, and $H \times W$ the image resolution.

- Mask statistics: $O(n_{\text{all}}HW)$.
- Masked track sets: $O(\sum_v q_v)$.
- View-level dominance filter: $O(n_{\text{all}}^2)$.
- Subset scoring: $O(\binom{n}{k}k^2\bar{q})$. The pairwise terms are $O(k^2)$ per subset, but $O(S)$ compares track sets rather than scalars, so each pair carries a factor $\bar{q}$.
- Exact Pareto filtering: $O(\binom{n}{k}^2)$ by all-pairs dominance. This term dominates the selector.
- Direct Chebyshev minimization over all subsets: $O(\binom{n}{k})$, which returns the same $S^\star$ (Section 2.3).
- Multiview filtering: $O(k(N_G + HW))$, one rasterization per selected view rather than k scalar projections per Gaussian.
- Export: $O(N_K)$ for the $N_K \leq N_G$ retained rows.

At the $k = 3$ operating point, candidate reduction keeps the exact front practical. Across the 18 evaluated scenes, the median reduced

Table 3: End-to-end cost of CleanGS over 18 scenes on one A100-80GB GPU, run sequentially. Selection runs on the CPU.

Stage	Median	Range
Selection, explicit front (main)	21.0 s	2.5–101.9 s
mask scoring, COLMAP, view data	∼20 s	2.4–50 s
explicit Pareto front	< 0.1 s	up to 51 s
Selection, direct argmin	17.3 s	2.5–50.3 s
Filtering and export	18.7 s	10.2–57.1 s
Total, explicit front	41 s	12.8–128 s
Total, direct argmin	40 s	12.8–77 s
Peak GPU memory, filtering	1.9 GB	0.9–6.1 GB
Peak CPU memory, selector / filtering	165 MB / 2.0 GB	

Table 4: Footprint-aware mask membership with the same three selected masks over 18 scenes. Std: captured framing; 6×: widened framing. p compares each row with center lookup (two-sided Wilcoxon).

Membership, vote	Leak (std)	PSNR (std)	Leak (6×)	PSNR (6×)	Retained
Center, 2-of-3 (main)	0.206	19.03	0.066	17.83	22.8%
$\tau = 0.5$, 2-of-3	0.194	19.03	0.064	17.85	22.9%
$\tau = 0.75$, 2-of-3	0.169	18.56	0.052	17.70	22.5%
$\tau = 0.9$, 2-of-3	0.148	17.96	0.044	17.32	22.1%
$\tau = 0.9$, 3-of-3	0.085	15.45	0.024	15.26	17.7%
$\tau = 0.9$, 3-of-4	0.109	16.15	0.026	15.72	19.3%
$\tau = 0.9$, 3-of-5	0.130	17.97	0.032	17.40	21.6%
$\tau = 0.75$, 3-of-5	0.150	18.56	0.039	17.77	22.0%

pool has $n = 10.5$ views, giving 142.5 scored subsets and a median Pareto front of 13, compared with the $\binom{60}{3} = 34{,}220$ subsets allowed by the pool cap. The reduction itself keeps the subset count small on all but 2 scenes. For dense reconstructions, the $\bar{q}$ factor in the masked support objective can dominate subset-scoring time.

The exact Pareto front makes the trade-offs among the four objectives explicit and provides a compact set of non-dominated alternatives that can be rescalarized under different weight vectors. This supports the weight-sensitivity analysis in Section 12 while preserving the multi-objective structure of the selector.

3.1 Measured end-to-end cost

Table 3 reports the cost of the complete CleanGS pipeline, measured sequentially on one A100-80GB GPU. The main paper's filtering-and-export time excludes view selection; including it, the median end-to-end time is 41 s (12.8–128 s). The selector runs on the CPU with a peak resident memory of 165 MB, and its time is dominated by reading and scoring masks rather than by the Pareto front, except on the two scenes whose pool reaches the cap. With the direct argmin, the end-to-end median is 40 s (12.8–77 s).

4 FILTERING AND METRIC DEFINITIONS

Renderer visibility. The indicator $\mathrm{vis}_v(g_i)$ is read directly from the gsplat rasterization of the full scene at camera v. A Gaussian counts as visible when the screen-space radius reported by the rasterizer is strictly positive, that is, it survives frustum culling and is assigned to at least one tile, and when its projected center $\pi_v(\mu_i)$, truncated to integer pixel coordinates, falls inside the image. Mask membership is then the value of the binary mask at that pixel. Note that this is a contribution test, instead of an occlusion test: a Gaussian lying behind the target is still visible in this sense, which is the ambiguity discussed in the main paper.

Masked SSIM. We compute the full-image SSIM map between render and reference with an 11×11 Gaussian window and data range 1, average it over the three colour channels, and take its mask-weighted mean, $\sum_p M_v(p)\,\mathrm{SSIM}(p) \,/\, \sum_p M_v(p)$. Windows that straddle the silhouette therefore include some surrounding context, as they would for any windowed structural measure.

Masked LPIPS. We crop both images to the tight bounding box of M_v, zero the pixels outside the mask within that crop, pad to at least 64×64 when the box is smaller, and evaluate AlexNet LPIPS on the result. Cropping keeps the comparison on the target rather than on a largely empty frame, which matters at 6× widening, where the target covers a small fraction of the image.

Widened-field scoring. At the widened framing, the reference RGB image and the target mask undergo the same image-plane transformation as the camera. Reference radiance exists only inside the original captured field of view, so masked PSNR, SSIM, and LPIPS are computed only over target pixels with captured reference RGB. Leakage, in contrast, is computed over all off-target pixels, including regions newly exposed by widening, because it requires only the rendered opacity.

Leakage. All reported leakage values are the unthresholded mean rendered opacity over off-target pixels,

$$\sum_p (1 - M_v(p)) A_v(p) / \sum_p (1 - M_v(p)),$$

averaged over evaluation views and then over scenes. The evaluator also computes the fraction of off-target pixels with $A_v(p) > 0.1$. At the captured framing this thresholded variant is 1.000 for the uncleaned scene, 0.201 for FlashSplat, 0.128 for GaussianCut, 0.084 for GaussianCut†, and 0.269 for CleanGS, preserving the ranking of the unthresholded measure.

5 FOOTPRINT-AWARE MASK MEMBERSHIP

The main filtering rule tests mask membership at the projected Gaussian center. Large anisotropic Gaussians can have screen-space footprints that extend beyond the mask even when their centers lie inside it. We therefore evaluate a footprint-aware variant: a visible Gaussian supports a view only if at least a fraction τ of its projected 2D Gaussian mass lies inside the mask. The mass is estimated from 64 fixed samples of the projected covariance, and samples that fall outside the image count as outside the mask. The vote and every other stage are unchanged. Table 4 reports the results with the same selected masks.

At $\tau = 0.5$, the footprint test lowers leakage from 0.206 to 0.194 ($p = 0.009$) while standard PSNR stays at 19.03 dB, so it directly reduces boundary leakage caused by wide footprints. Stricter thresholds provide a controlled fidelity–cleanup knob: $\tau = 0.75$ and $\tau = 0.9$ lower leakage to 0.169 and 0.148 at a cost of 0.48 and 1.08 dB in

standard PSNR (both $p < 0.001$). Gaussian counts decrease monotonically with τ on all 18 scenes. Section 11 places these operating points on the fidelity–cleanup frontier.

6 PER-SCENE RESULTS

Tables 5 and 6 report per-scene photometric results at the captured and widened framings, respectively, while Table 7 reports cleanup diagnostics. The GaussianCut columns correspond to the all-mask GaussianCut† configuration from the main paper. The FlashSplat columns use the same three selected masks as CleanGS. Photometric macro averages match those reported in the main paper; leakage in Table 7 is measured under widened framing.

Tables 5 and 6 show that the aggregate improvements are consistent across scenes. At both framings, CleanGS achieves higher PSNR than FlashSplat on 16 of 18 scenes and than all-mask GaussianCut† on 15 of 18, and outperforms matched three-mask GaussianCut on all 18 scenes. Its mean PSNR is 19.03 dB at the captured framing and 17.83 dB after widening, with per-scene standard deviations of 2.83 and 2.93 dB, respectively.

The comparison with the uncleaned reconstruction shows the intended trade-off more clearly. CleanGS improves PSNR on 9 of 18 scenes at the captured framing, but on 16 of 18 under widened evaluation. This mirrors the aggregate results: cleanup largely preserves the appearance of the original reconstruction in the captured views while substantially improving robustness to reframing.

Median PSNR. Table 8 reports median alongside mean PSNR. The ranking of the methods is unchanged at both framings. Medians are less sensitive to individual scenes such as *super markets*, where GaussianCut† reaches only 4.26 dB. The mean margin of CleanGS over GaussianCut† is 1.76/1.12 dB (standard/widened) over all 18 scenes and 1.49/0.95 dB when *super markets* is excluded.

7 CHOICE OF ZOOM FACTOR

The widened protocol reduces focal length at a fixed pose and a fixed output resolution, so the target occupies fewer pixels and the masked metrics are computed over a smaller region. Averaged over the 18 scenes, the evaluated target pixel count Z_v is 5.8×10^5 at the captured framing and 2.3×10^4 at 6×, a ratio of 24.6. Pure area scaling would predict 36; the difference is that several targets are already clipped by the captured frame, so Z_v at 1× is smaller than the full silhouette would imply.

The widened-field protocol divides focal length by six. Table 9 repeats it at 2× and 4× on the same 18 scenes and the same exported assets, so the only thing that varies is the focal length.

CleanGS ranks first at every widened factor, while FlashSplat and GaussianCut keep the same relative order at 2×, 4×, and 6×. The result is therefore stable to the chosen widening factor. The uncleaned scene follows the complementary trend: it leads at captured framing, is overtaken by 2×, and ranks last by 4× as retained off-object structure progressively enters the widened view.

Table 10 reports leakage at each widening factor. Leakage falls with widening for every method because the target and the retained structure near it occupy a smaller share of the widened image while newly exposed regions are mostly empty. The ranking is stable across factors. GaussianCut† has the lowest leakage at every factor, followed by matched GaussianCut and FlashSplat, while CleanGS has the highest leakage among the cleanup methods at every factor, consistent with its appearance-preserving operating point (Section 11).

8 BASELINE EVALUATION SETUP

FlashSplat. We evaluate FlashSplat using its exported object with removal slackness 0, the default setting of the reference implementation. It receives the same three selected masks as CleanGS's 3D filtering stage and reaches 16.44/16.29 dB standard/widened PSNR at leakage 0.141 and 14.9% retention. Its lower leakage does not come from empty outputs: the mean rendered opacity inside the target mask is 0.94, against 0.96 for CleanGS, and no scene falls below 0.86. The PSNR gap instead comes from target surfaces that are partly removed or smeared, most visibly on *Francis*, *garbage*, and *picnic*.

The removal-slackness sweep in Section 11 was run on a different three-view subset of the same curated masks, where FlashSplat at slackness 0 reaches 17.34/16.79 dB, leakage 0.148, and 17.4% retention. On that subset, changing the slackness to −0.5 increases retention to 61.7% and leakage to 0.446, which shows how sensitive FlashSplat's retained asset is to this parameter.

GaussianCut. We evaluate the final graph-cut output under two supervision settings. The matched configuration receives the same three Pareto-selected masks as CleanGS. The second configuration, denoted GaussianCut†, receives all available capture masks, including masks at scored cameras, and serves as an all-mask reference. Both configurations are evaluated with the same protocol over the same 18 scenes.

Additional methods. We restrict the quantitative comparison to methods that produce a directly renderable Gaussian subset under comparable preprocessing. LUDVIG [5] applies a fixed Gaussian-budget pruning stage before semantic uplifting, so its output is not directly comparable to cleanup applied to the original pretrained scene. SAGOnline [10] is not included because a directly comparable retained-asset result is not available under this evaluation protocol. Gaussian Grouping and Gaga represent object identity through learned or rasterized semantic labels rather than an exported pruned Gaussian subset; in our Gaga evaluation, the exported scene retains all input Gaussians. We therefore discuss these methods as related approaches rather than include them in the retained-asset table.

Recent Gaussian-level segmentation methods also lift multi-view 2D masks to Gaussians, but under a different setting. Click-Gaussian [1] trains contrastive per-Gaussian features from dense SAM masks of all training views. Trace3D [9] densifies and prunes Gaussians during training to obtain consistent instance segmentation. SAGOnline [10] propagates masks across rendered views with a video segmenter and fuses per-Gaussian labels, with per-scene refinement. Specifically, none of them exports a subset of the original pretrained Gaussians from three masks without retraining. So, we treat them as related work rather than as baselines.

Masks. Target masks were produced once during data curation with SAM 3 and manual prompts, checked only for coverage of the intended object, and reused unchanged in every run and for every method. Importantly, no mask was edited for any particular method. CleanGS, matched GaussianCut, and FlashSplat use the same three

Table 5: Per-scene standard-view photometrics. Results over the same 18 scenes as the main comparison.

Dataset	Scene	Frames	Uncleaned			FlashSplat			GaussianCut†			CleanGS		
			PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
DL3DV	car_set2	43	18.61	0.704	0.225	17.51	0.681	0.249	17.85	0.687	0.241	18.39	0.715	0.204
	courthouses	52	19.37	0.665	0.239	18.90	0.648	0.252	18.97	0.650	0.249	22.89	0.703	0.202
	fountain	44	20.63	0.602	0.096	14.69	0.487	0.215	18.19	0.566	0.143	19.47	0.632	0.108
	gardens	44	18.79	0.551	0.153	15.84	0.505	0.208	17.45	0.537	0.170	18.64	0.579	0.143
	red_brick_build	39	20.05	0.566	0.237	16.45	0.514	0.293	17.13	0.529	0.277	16.34	0.541	0.278
	redcar	41	18.24	0.688	0.222	17.29	0.669	0.247	17.57	0.673	0.239	19.15	0.725	0.184
	super_markets	47	10.91	0.396	0.579	7.50	0.322	0.659	4.26	0.179	0.788	10.76	0.412	0.565
	tractor	41	20.66	0.736	0.133	18.40	0.683	0.170	17.13	0.675	0.184	19.91	0.760	0.130
NeRFBusters	car	42	17.88	0.616	0.244	16.67	0.570	0.300	16.84	0.579	0.287	19.00	0.650	0.234
	century	50	19.71	0.686	0.126	15.20	0.529	0.258	17.18	0.628	0.193	19.94	0.741	0.109
	flowers	44	19.81	0.533	0.151	15.50	0.441	0.202	15.39	0.439	0.206	16.39	0.521	0.171
	garbage	45	19.77	0.717	0.211	14.08	0.611	0.351	19.32	0.708	0.215	22.06	0.768	0.165
	picnic	40	19.29	0.508	0.090	14.07	0.399	0.191	18.35	0.475	0.112	16.37	0.464	0.149
	roses	46	19.46	0.519	0.105	18.01	0.492	0.138	18.93	0.504	0.117	20.06	0.593	0.081
	table	36	26.11	0.909	0.076	24.42	0.900	0.083	20.03	0.774	0.143	19.94	0.880	0.105
TnT	Family	42	21.17	0.606	0.262	20.12	0.584	0.273	20.32	0.588	0.272	23.80	0.689	0.230
	Francis	42	18.44	0.510	0.192	14.38	0.415	0.340	17.32	0.481	0.210	18.60	0.499	0.188
	Horse	42	19.39	0.647	0.158	16.91	0.578	0.203	18.66	0.624	0.171	20.89	0.707	0.144

Table 6: Per-scene widened-field photometrics. Results over the same 18 scenes at 6× widened framing.

Dataset	Scene	Frames	Uncleaned			FlashSplat			GaussianCut†			CleanGS		
			PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
DL3DV	car_set2	43	15.31	0.445	0.313	17.50	0.508	0.211	17.78	0.508	0.196	18.45	0.577	0.169
	courthouses	52	12.22	0.324	0.492	17.65	0.480	0.238	17.59	0.480	0.237	20.78	0.568	0.184
	fountain	44	12.64	0.192	0.214	16.01	0.430	0.091	18.60	0.489	0.061	20.14	0.604	0.048
	gardens	44	12.32	0.159	0.219	16.38	0.414	0.096	16.77	0.428	0.089	17.22	0.484	0.088
	red_brick_build	39	19.54	0.353	0.295	16.55	0.315	0.324	17.16	0.331	0.313	16.50	0.337	0.312
	redcar	41	12.77	0.359	0.419	17.25	0.482	0.198	17.45	0.484	0.190	18.99	0.592	0.145
	super_markets	47	6.22	0.109	0.807	7.28	0.223	0.643	4.25	0.112	0.758	8.30	0.275	0.609
	tractor	41	9.25	0.208	0.382	17.64	0.513	0.122	16.58	0.492	0.145	18.71	0.650	0.094
NeRFBusters	car	42	12.66	0.317	0.301	15.60	0.394	0.212	15.71	0.397	0.201	16.57	0.471	0.201
	century	50	15.98	0.366	0.116	15.04	0.331	0.120	15.72	0.349	0.112	16.54	0.393	0.106
	flowers	44	14.06	0.240	0.288	15.80	0.292	0.187	15.66	0.280	0.190	16.21	0.330	0.184
	garbage	45	12.20	0.439	0.342	14.04	0.437	0.311	18.39	0.578	0.182	19.84	0.645	0.159
	picnic	40	16.32	0.356	0.104	13.87	0.321	0.140	16.50	0.355	0.094	15.01	0.332	0.115
	roses	46	15.29	0.249	0.240	16.89	0.323	0.096	17.13	0.318	0.096	18.22	0.420	0.074
	table	36	13.28	0.687	0.143	24.48	0.773	0.071	20.38	0.642	0.151	20.48	0.728	0.093
TnT	Family	42	11.31	0.185	0.457	19.33	0.485	0.172	19.44	0.498	0.159	22.15	0.629	0.128
	Francis	42	12.99	0.253	0.427	14.71	0.287	0.318	17.55	0.393	0.189	18.90	0.429	0.174
	Horse	42	6.69	0.150	0.333	17.18	0.388	0.137	18.06	0.447	0.119	17.96	0.555	0.119

selected training-view masks. Evaluation masks are never available to the selector or to the scalarization.

9 SELECTED VIEWS AGAINST RANDOM VIEWS

Table 11 expands the selector comparison from the main paper. Both arms use the same two-of-three vote and evaluator; only the three input views differ. The random arm samples three training views uniformly with five seeds per scene and averages the seeds within each scene before pairing.

Across the 17 scenes with defined photometrics in both arms, selected views improve all six image-quality metrics: +1.56 dB standard PSNR and +1.00 dB widened PSNR, together with corresponding SSIM and LPIPS gains. Random subsets produce lower leakage (0.154 versus 0.197) and lower retention (18.8% versus 21.7%); these compactness differences are also significant ($p = 0.0093$ and $p = 0.0067$, respectively). The comparison therefore shows a clear operating-point shift: selected views preserve more of the target while random views prune more aggressively.

Table 7: Per-scene cleanup diagnostics. Widened-field silhouette leakage and retained Gaussian fraction.

		Leakage ↓			Retained ↓		
Dataset	Scene	FS	GCut†	Ours	FS	GCut†	Ours
DL3DV	car_set2	0.023	0.012	0.037	25.8%	32.9%	32.6%
	courthouses	0.023	0.012	0.037	7.2%	8.2%	7.6%
	fountain	0.046	0.000	0.047	1.1%	1.5%	1.8%
	gardens	0.006	0.000	0.082	2.5%	3.7%	5.8%
	red_brick_build	0.019	0.026	0.016	29.2%	41.9%	29.3%
	redcar	0.008	0.001	0.020	24.2%	28.3%	31.6%
	super_markets	0.098	0.003	0.221	11.3%	4.6%	42.7%
	tractor	0.002	0.001	0.017	18.0%	20.4%	21.0%
NeRFBusters	car	0.062	0.013	0.105	3.7%	4.7%	9.0%
	century	0.131	0.040	0.125	22.2%	33.3%	44.9%
	flowers	0.076	0.003	0.038	30.8%	33.0%	31.9%
	garbage	0.029	0.013	0.129	6.6%	15.8%	17.0%
	picnic	0.035	0.012	0.099	4.4%	9.3%	15.7%
	roses	0.026	0.001	0.023	31.8%	40.0%	38.6%
	table	0.008	0.004	0.008	6.9%	9.0%	6.2%
TnT	Family	0.116	0.002	0.013	9.3%	8.7%	10.0%
	Francis	0.037	0.017	0.095	26.6%	57.9%	53.3%
	Horse	0.003	0.000	0.081	7.1%	10.8%	12.1%

Table 8: Mean and median masked PSNR over 18 scenes. Std: captured framing; 6×: widened framing.

	Mean		Median	
Method	Std	6×	Std	6×
Uncleaned 3DGS	**19.35**	12.84	**19.43**	12.71
FlashSplat	16.44	16.29	16.56	16.47
GaussianCut	15.28	15.10	16.40	16.02
GaussianCut†	17.27	16.71	17.71	17.30
CleanGS	19.03	**17.83**	19.31	**18.33**

Table 9: PSNR across widening factors. Masked PSNR at captured framing and at 2×, 4×, and 6× widening over the same 18 exported assets. CleanGS ranks first at each tested widened setting.

Method	1×	2×	4×	6×
Uncleaned 3DGS	**19.35**	18.72	14.72	12.84
FlashSplat	16.44	16.65	16.59	16.29
GaussianCut†	17.27	17.43	17.15	16.71
CleanGS	19.03	**19.31**	**18.73**	**17.83**

On *super markets*, three of the five random subsets retain no Gaussians and therefore have undefined photometrics, whereas the selected subset remains non-empty. Our selected main configuration produces a non-empty asset on all 18 scenes.

Do the selected views help other filters? Table 12 applies the same selected masks and the first random three-view set (seed 0) to GaussianCut and FlashSplat. Neither baseline benefits from the selected views. In contrast, CleanGS gains 0.80 dB in standard PSNR from selection ($p = 0.017$), while its leakage and retention rise. The selected masks are therefore not uniformly better inputs for every lifting method. They are useful specifically for the vote, where reliable views make agreement meaningful. Selection also improves robustness: with random views, a strict three-of-three vote empties *car* and *super markets* for two of three seeds, whereas the selected views never produce an empty asset.

Table 10: Leakage across widening factors. Mean unthresholded silhouette leakage over the same 18 exported assets at captured framing (1×) and at 2×, 4×, and 6× widening.

Method	1×	2×	4×	6×
Uncleaned 3DGS	0.983	0.896	0.813	0.803
FlashSplat	0.141	0.085	0.049	0.042
GaussianCut	0.089	0.047	0.022	0.016
GaussianCut†	**0.055**	**0.038**	**0.015**	**0.009**
CleanGS	0.206	0.123	0.075	0.066

Table 11: Pareto-selected views against random views at the same budget. Both arms use the same vote and evaluator and differ only in the three input views. Five seeds are averaged within each scene, with paired results reported over the 17 scenes with defined outputs for both arms. "Wins" counts scenes favoring selection, and p is a two-sided Wilcoxon signed-rank test. Std: captured framing; 6×: widened framing.

	Pareto	Random	Δ	Wins	p
PSNR ↑	**19.52**	17.96	+1.56	15/17	0.0005
SSIM ↑	**0.657**	0.620	+0.037	15/17	0.0004
LPIPS ↓	**0.166**	0.202	−0.036	15/17	0.0011
PSNR (6×) ↑	**18.39**	17.39	+1.00	14/17	0.0150
SSIM (6×) ↑	**0.514**	0.481	+0.033	14/17	0.0038
LPIPS (6×) ↓	**0.141**	0.170	−0.029	13/17	0.0129
Leakage ↓	0.197	0.154	+0.043	5/17	0.0093
Retained ↓	21.7%	18.8%	+2.9	4/17	0.0067

Table 12: Selected against random masks for each filter. Each filter receives either the three selected masks or the seed-0 random three-view set. Std: captured framing; 6×: widened framing. p is a two-sided Wilcoxon test on standard PSNR. The CleanGS rows cover 17 scenes because the random set empties *super markets*.

Filter	Masks	PSNR (std)	PSNR (6×)	Leak	Retained	p
GaussianCut	selected	15.28	15.10	0.089	17.2%	0.20
	random	15.92	15.72	0.085	18.2%	
FlashSplat	selected	16.44	16.29	0.141	14.9%	0.15
	random	16.91	16.56	0.149	16.0%	
CleanGS	selected	19.52	18.39	0.197	21.7%	0.017
	random	18.72	18.02	0.158	19.3%	

Camera variation. Because the selector uses camera baseline and viewing-direction diversity, its benefit could depend on how much the capture trajectory varies. We measure the angular spread of each scene as the mean pairwise viewing-direction angle over all training cameras with candidate masks, and the selection gain as the paired PSNR difference between selected and random views from Table 11. The gain does not follow the spread (Spearman $\rho = 0.24$ at captured framing and 0.13 widened). The only scene where selection loses at both framings is *red brick build* ($-0.67/-0.70$ dB), which also has the narrowest spread (31°).

10 ROLE OF THE PARETO FRONT

The dominance lemma guarantees that any coordinate-wise non-decreasing utility attains an optimum on the exact Pareto front of the enumerated subsets of $\mathcal{V}_r$. Restricting selection to that front therefore preserves the optimal value of the weighted Chebyshev criterion over the search space the selector actually uses, while keeping the trade-offs among A, B, O, and T explicit.

This is particularly useful in our setting because the goal is not to collect as many observations as possible, but to identify a small subset that provides strong and complementary evidence. With only three selected views, redundancy is costly: geometrically or semantically similar observations spend the limited view budget repeating much of the same evidence. The multi-objective formulation instead allocates this budget deliberately across mask reliability, camera baseline, viewing-angle diversity, and masked-support complementarity before a single operating point is chosen. The comparison with random three-view subsets in Table 11 shows the practical benefit of selecting the three-view evidence rather than choosing it arbitrarily.

The front also keeps the non-dominated trade-offs available when the relative importance of the objectives changes. Table 16 shows that reweighting the same objective space moves the selected operating point while maintaining similar overall fidelity. Exact Pareto construction becomes more expensive as k increases because the number of feasible subsets grows combinatorially, as reflected in Table 18. We therefore use $k = 3$, where exact Pareto selection remains practical while providing the redundancy required for two-of-three voting. Because the direct argmin returns the same subset (Section 2.3), larger budgets do not require the explicit front.

Pareto selection against simpler selectors. Table 13 compares the Pareto selector with four simpler selectors that use the same reduced pool, vote, and evaluator: a weighted sum of the same normalized objectives and weights, a greedy selector maximizing the same weighted objective, the three most reliable views, and the most reliable view followed by farthest-point sampling of camera centers. They choose a different subset from Pareto selection on 11, 16, 11, and 11 scenes, respectively. All four stay within 0.08 dB of Pareto selection in standard PSNR. Pareto selection has lower leakage than top-three reliability ($p = 0.016$), whereas reliability with farthest-point sampling has higher widened PSNR ($p = 0.044$) and lower leakage ($p = 0.001$) than Pareto selection on these scenes. It is worth noting however that the Pareto formulation adds an explicit, tunable trade-off among reliability, baseline, and support, instead of a single hard-coded heuristic, which, as we show, does not generalize well.

Table 13: Pareto selection against simpler selectors on 18 scenes, with the same reduced pool, two-of-three vote, and evaluator. Std: captured framing; 6×: widened framing.

Selector	PSNR (std)	PSNR (6×)	Leak (std)	Leak (6×)	Retained
Pareto (main)	19.03	17.83	0.206	0.066	22.8%
Weighted sum	19.00	17.71	0.193	0.060	21.3%
Greedy	18.96	17.72	0.194	0.060	21.5%
Top-3 reliability	19.07	17.70	0.224	0.077	23.1%
Reliability + farthest point	19.02	18.15	0.166	0.032	20.6%

As an exploratory check outside the 18 benchmark scenes, we ran the selectors on 15 further scenes from the same benchmarks that have all required inputs. There, reliability with farthest-point sampling nearly empties one scene (*m60*: 1,111 retained Gaussians and 6.0 dB, against 539,557 Gaussians for Pareto selection), because it ignores masked support. Replacing the mean pairwise baseline in $B(S)$ by the smallest pairwise baseline (a max–min form) changes the selection on 9 of 15 scenes and raises standard PSNR by 0.13 dB ($p = 0.012$) by removing near-duplicate views. We report this as an indication of how the objectives can be tuned.

Geometry and complementarity of the selected views. Selected views are geometrically closer to one another than random triplets: their median smallest pairwise viewing angle is 7.6° against 34° for random views ($p = 0.014$), and in 10 of 18 scenes two of the three selected views lie within 10°. This follows from the weighting, in which the weakest-link reliability A dominates and B and T are normalized within a small reduced pool. In terms of target evidence, the selected views remain complementary in absolute terms: in the median scene, 90% of the in-mask SfM points are seen by only one of the three views, and whichever view is added third still contributes at least 22% new points. They are not more complementary than random triplets, however (median O of 0.943 against 0.987), and *Francis*, *garbage*, and *picnic* are near-redundant. We therefore describe the selected views as reliable views that each contribute non-redundant target evidence.

11 FIDELITY–CLEANUP FRONTIER

Masked PSNR rewards keeping the target, while leakage and retention reward pruning, so a single operating point cannot show whether a method improves the fidelity–cleanup trade-off or only moves along it. Figure 1 therefore plots PSNR against leakage. With the same three selected masks, we vary CleanGS through the vote threshold (one-, two-, or three-of-three) and the footprint threshold τ (Section 5); hollow circles use five selected masks. We vary FlashSplat through its removal slackness, and run random-view CleanGS with the same filters on random three-view subsets (seeds 0–2). All points are means over the 18 scenes, and the $k \geq 4$ selections use the direct argmin.

At captured framing and the same three-mask budget, the CleanGS curve lies above every matched-budget baseline point. At the leakage of FlashSplat's slackness-0 point on the sweep subset (0.148), it reaches 17.96 dB, 0.62 dB higher, and FlashSplat with our selected masks (0.141, 16.44 dB) lies further below the curve. At low leakage, $\tau = 0.9$ with a three-of-three vote dominates matched GaussianCut (0.085/15.45 against 0.089/15.28). Only the

Table 14: Sensitivity to the occupancy width σ with minimum aggregation over 18 scenes; all other settings are fixed. "Changed" counts the scenes whose three selected views differ from those at the main $\sigma = 0.15$; the other scenes keep the main result. Std: captured framing; 6×: widened framing.

σ	Changed	PSNR (std)	PSNR (6×)	Leak (std)	Leak (6×)
0.05	5/18	19.14	17.76	0.206	0.073
0.10	1/18	19.05	17.84	0.207	0.066
0.15 (main)	–	19.03	17.83	0.206	0.066
0.20	2/18	19.05	17.70	0.205	0.072
0.30	3/18	19.04	17.96	0.206	0.065

all-mask GaussianCut†, which uses masks at the scored cameras, stays above the curve at low leakage. Under widened framing, $\tau = 0.9$ with two-of-three voting has lower leakage (0.044) and at least 0.5 dB higher PSNR than every FlashSplat slackness setting, and FlashSplat with our selected masks is 1.0 dB lower at similar leakage (0.042). With five masks, three-of-five voting matches the standard PSNR of the main configuration (19.03 dB) at lower leakage (0.180 against 0.206), and with $\tau = 0.9$ it reaches 17.97 dB at 0.130.

The random-view curve isolates the contribution of selection under the same filter. Selected views are 0.6–1.1 dB higher than random views for leakage between 0.15 and 0.35, while below about 0.13 random views are up to 0.3 dB higher; the curves cross near 0.14. Selection therefore lifts the frontier in the moderate-leakage region that contains the main operating point, rather than only moving to a more retentive point, while under very aggressive cleanup the difference is mainly a shift in operating point. Paired on the same footprint filter, selection still raises PSNR together with leakage (+1.54 dB at $\tau = 0.9$, $p = 0.001$).

12 SELECTOR ABLATIONS

The following ablations vary one component of the selector while keeping the remaining pipeline fixed. Each arm is paired against a rerun of the main k=3, 2-of-3 configuration inside the same ablation family. That reference covers 17 of the 18 benchmark scenes; the NeRFBusters *car* scene is excluded because its ablation runs did not complete across the arms. Its leakage on those 17 scenes is 0.204, measured at the captured (1×) framing on the full evaluation masks, the same protocol as Table 1 (Main paper), whose 0.206 is the corresponding 18-scene mean. Arms that did not complete on all 17 scenes are reported on their own completed subsets.

Reliability aggregation. The reliability objective $A(S)$ can aggregate the per-view scores of a subset by their minimum, as in the main paper, or by their mean. We report the ablations for both settings. Tables 15–18 use the mean, and each arm is paired with a reference run that also uses the mean. Tables 14 and 19 use the minimum and report the sensitivity to the occupancy width σ and the budget and threshold study on all 18 scenes.

Across $\sigma \in [0.05, 0.30]$, the selected views differ from the main selection on at most 5 of 18 scenes and mean standard PSNR changes by at most 0.11 dB, so the selection is stable over this range.

Table 15: Sensitivity to zeroing individual scalarization weights. Each row sets one Chebyshev weight to zero and renormalizes the remaining weights while retaining the four-objective Pareto front, thereby measuring the influence of each weight. Mean aggregation. Std: captured framing; 6×: widened framing.

Weight zeroed	ΔPSNR	ΔPSNR$_{6\times}$	Median $\Delta_{6\times}$	Leakage ↓
Main	–	–	–	0.204
without A	-0.48	-0.07	+0.01	0.163
without B	+0.13	+0.09	+0.00	0.204
without O	-0.08	-0.16	-0.05	0.218
without T	-0.05	-0.22	-0.07	0.211

Table 16: Chebyshev weight sensitivity. The main weights are $(A, B, O, T) = (0.45, 0.30, 0.15, 0.10)$. Each variant changes the relative emphasis placed on the four objectives. Mean aggregation. Std: captured framing; 6×: widened framing.

Weights	n	ΔPSNR	ΔPSNR$_{6\times}$	Med. $\Delta_{6\times}$	Leak ↓	Wins
Main	17	–	–	–	0.204	–
equal, 0.25 each	17	+0.02	−0.03	+0.00	0.192	5/17
B heavy, 0.40	17	+0.14	+0.17	+0.08	0.188	9/17
O heavy, 0.40	17	−0.02	+0.07	+0.00	0.188	5/17
T heavy, 0.40	17	+0.13	+0.06	+0.00	0.191	7/17
O, T heavy, 0.35	17	+0.09	+0.48	+0.00	0.180	8/17

Table 15 examines how much each scalarization weight influences selection at the main weighting. The Pareto front is unchanged in every row, so these are weight ablations and not objective ablations. Zeroing the weight on A reduces standard PSNR by 0.48 dB and changes the selected subset on 14 of 17 scenes, making mask reliability the strongest individual contributor at this operating point. Zeroing B changes 10 of 17 selections, with a small +0.13 dB change in standard PSNR. Zeroing O or T carries a smaller but consistent cost: standard PSNR falls by 0.08 and 0.05 dB and widened PSNR by 0.16 and 0.22 dB, with negative medians in both cases, while leakage rises from 0.204 to 0.218 and 0.211, respectively. The diversity terms therefore contribute least at the main weighting; their effect is largest under widened evaluation, where redundant supporting views are most exposed.

The main weighting deliberately emphasizes A and B. Mask reliability directly controls the quality of the semantic evidence entering the vote, while camera baseline provides a strong geometric cue for avoiding redundant observations. The support-overlap and viewing-angle objectives provide additional notions of diversity, but receive smaller weights because their informativeness depends more strongly on the available sparse tracks and capture trajectory.

Table 16 shows that performance remains stable across a broad range of weightings: standard PSNR varies by at most 0.14 dB across the tested settings. Increasing the emphasis on O and T changes the selected subsets and shifts the macro operating point under widened evaluation: the overlap-and-angle-heavy setting improves widened PSNR by 0.48 dB and lowers leakage from 0.204 to 0.180.

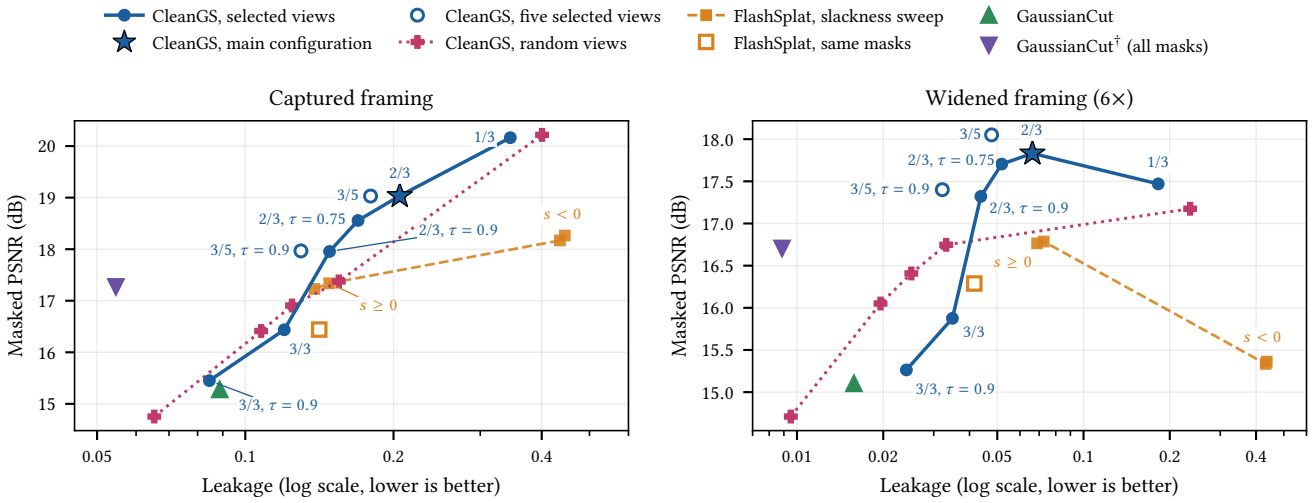


Figure 1: Leakage–PSNR frontier over 18 scenes at captured (left) and 6× widened (right) framing, with leakage on a log scale. Labels give the vote (k-of-n) and the footprint threshold τ. The FlashSplat sweep varies the removal slackness s on a different three-view subset; the hollow square is FlashSplat at $s = 0$ with our selected masks. Random-view points use the same filters and average seeds 0–2.

Table 17: Per-view reliability score. The main configuration uses a Gaussian-shaped occupancy score with r_0=0.30 and σ=0.15; each variant changes the occupancy target or decay profile while leaving the remaining pipeline fixed. Mean aggregation. Std: captured framing; 6×: widened framing.

Reliability score	n	ΔPSNR	ΔPSNR$_{6\times}$	Med. $\Delta_{6\times}$	Leak ↓	Wins
Main	17	–	–	–	0.204	–
target 0.05	17	−1.69	−1.43	−0.18	0.164	5/17
target 0.75	17	−0.18	−0.20	+0.00	0.215	4/17
linear, target 1.0	17	−0.27	−0.26	+0.00	0.215	4/17
linear decay	17	−0.01	+0.39	+0.00	0.183	7/17
step decay	17	+0.19	+0.56	+0.18	0.155	9/17

That gain is not uniform across scenes, however: the same arm improves widened PSNR on 8 of the 17 scenes, with a median per-scene change of 0.00 dB. Thus, A and B determine most selections at the main weights, while larger weights on O and T expose alternative operating points with different fidelity–cleanup trade-offs, indicating room for further tuning.

Table 17 examines the per-view reliability score. Moving the occupancy target to $r_0 = 0.05$ reduces standard PSNR by 1.69 dB, while favoring very large masks through $r_0 = 0.75$ or raw-area scoring reduces it by 0.18–0.27 dB. The degradation at both extremes supports the band-pass design: useful views should contain enough target area to provide stable support without becoming near-full-frame close-ups.

Performance is less sensitive to the precise decay profile. Linear decay remains close to the Gaussian score, while a step profile improves the macro widened PSNR by 0.56 dB and lowers leakage, winning on 9 of the 17 scenes with a median per-scene change of +0.18 dB. The main benefit therefore comes from favoring an intermediate occupancy range, while the precise shape of the reliability function offers room for further refinement.

Table 18: Selected-view budget and vote threshold. Each arm is paired against the main k=3, 2-of-3 configuration on its completed scene set. "Median" is the median per-scene change in widened PSNR, and "Wins" counts scenes where the arm performs better. Exact Pareto construction is all-pairs dominance over $\binom{n}{k}$ subsets, which is 3.4×10^4 at k=3 but 4.9×10^5 at k=4 and 5.5×10^6 at k=5 for a capped pool of 60. Selection does not depend on the vote threshold, so the two k=4 rows differ in n for run-level resource reasons alone and are not directly comparable with each other. Mean aggregation. Std: captured framing; 6×: widened framing.

Configuration	n	ΔPSNR	ΔPSNR$_{6\times}$	Med. $\Delta_{6\times}$	Leak ↓	Wins
Main	17	–	–	–	0.204	–
k=2, 2-of-2	17	−2.28	−1.52	−0.18	0.115	6/17
k=4, 2-of-4	15	+0.52	+0.21	+0.04	0.220	10/15
k=5, 2-of-5	13	+0.57	−0.18	+0.01	0.250	9/13
k=4, 3-of-4	17	−1.19	−0.66	−0.03	0.144	8/17
k=5, 3-of-5	13	+0.10	+0.54	+0.13	0.147	10/13

Table 18 studies the interaction between view budget and vote threshold. The k=2 arm uses unanimity, so it changes both the view budget and the relative strictness of the vote; it represents the natural two-view operating point rather than a budget-only control. Relative to the main 2-of-3 configuration, it reduces standard and widened PSNR by 2.28 and 1.52 dB, respectively.

Table 19: Selected-view budget and vote threshold with minimum aggregation on all 18 scenes. Std: captured framing; 6×: widened framing.

Configuration	PSNR (std)	PSNR (6×)	Leak (std)	Leak (6×)	Retained
k=3, 1-of-3	20.16	17.47	0.345	0.182	29.1%
k=3, 2-of-3 (main)	19.03	17.83	0.206	0.066	22.8%
k=3, 3-of-3	16.44	15.88	0.120	0.035	18.6%
k=2, 2-of-2	16.87	16.25	0.132	0.036	19.3%
k=4, 3-of-4	16.96	16.19	0.155	0.040	20.1%
k=5, 3-of-5	19.03	18.05	0.180	0.048	22.4%

Table 20: Largest-component geometric refinement. Macro averages over 18 scenes. Both endpoints use the same eroded evaluation masks, isolating the effect of the clustering stage; their absolute values are therefore not directly comparable with the Table 1 (Main paper) endpoint, which is evaluated on the full masks. Std: captured framing; 6×: widened framing.

	Standard			Widened			Leak
Endpoint	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	↓
Vote only	**19.86**	**0.658**	**0.170**	**18.25**	**0.516**	**0.158**	0.221
+ clustering	16.02	0.478	0.281	14.97	0.370	0.303	<u>0.115</u>

Increasing the number of views while keeping a threshold of two provides only small median widened gains and increases leakage. Raising the threshold does not help uniformly: 3-of-4 loses 1.19 and 0.66 dB in standard and widened PSNR on its 17 scenes. The 3-of-5 configuration is the strongest larger-budget setting among the completed runs, improving standard PSNR by 0.10 dB and widened PSNR by 0.54 dB on its 13-scene subset while reducing leakage to 0.147.

Table 19 reports the same study with minimum aggregation on all 18 scenes. Arms with $k \geq 4$ use the direct argmin. Three-of-five voting matches the standard PSNR of the main configuration (−0.004 dB), improves widened PSNR by 0.22 dB, and lowers leakage at both framings (0.206 to 0.180 captured, 0.066 to 0.048 widened). On average, it is therefore a better operating point, although each arm wins on about half of the scenes. Raising the threshold at a fixed budget, or using unanimity at $k = 2$, lowers leakage at a substantial cost in fidelity.

The k=3, 2-of-3 configuration therefore provides a practical operating point: it is the smallest setting with voting redundancy and supports exact Pareto selection across all 18 scenes. The 3-of-5 result shows that additional views with a correspondingly higher threshold can improve the fidelity–cleanup trade-off.

13 PER-SCENE EFFECT OF CLUSTERING

Table 20 shows that spatial connectivity provides useful cleanup signal: retaining the largest HDBSCAN component reduces silhouette leakage from 0.221 to 0.115. The corresponding loss in photometric fidelity is highly scene dependent. Six scenes lose more than 3 dB in standard PSNR, while the remaining twelve lose at most 0.48 dB.

Table 21 shows that the largest fidelity losses are concentrated in scenes where the voted target is distributed across multiple density components. For example, the largest component contains only 9%, 13%, and 16% of the retained Gaussians on *tractor*, *redcar*, and *table*, respectively, so retaining a single component removes substantial target structure. In contrast, clustering has little effect on many scenes in which most of the retained target lies in one component.

These results show that connectivity can remove residual structure, but that component size alone is not a reliable criterion for identifying the target. A geometry-aware refinement should therefore use spatial structure together with semantic evidence rather than assume that the largest density component is the object.

Table 21: Per-scene effect of largest-component clustering. Scenes are ordered by standard-PSNR change. "Cl." is the number of HDBSCAN clusters in the voted set, and "top" is the fraction of voted Gaussians in the largest cluster. Values use the same eroded evaluation masks as Table 20. Std: captured framing; 6×: widened framing.

	Std PSNR			Leakage		Clustering	
Scene	vote	+cl.	Δ	vote	+cl.	Cl.	top
table	22.29	2.72	-19.57	0.226	0.012	112	16%
tractor	20.56	8.97	-11.58	0.223	0.002	92	9%
redcar	19.33	8.58	-10.75	0.272	0.004	39	13%
Horse	22.32	11.69	-10.63	0.070	0.005	22	47%
car_set2	18.56	8.50	-10.06	0.372	0.004	35	16%
red_brick_build	16.51	11.69	-4.81	0.200	0.031	7	88%
fountain	21.46	20.98	-0.48	0.132	0.019	34	74%
super_markets	10.84	10.56	-0.28	0.357	0.218	7	67%
gardens	19.90	19.67	-0.23	0.081	0.005	56	57%
Family	24.76	24.61	-0.16	0.062	0.038	2	96%
garbage	22.40	22.25	-0.15	0.345	0.308	3	97%
car	19.14	19.06	-0.08	0.224	0.193	4	98%
courthouses	23.36	23.29	-0.07	0.054	0.044	2	100%
flowers	17.20	17.14	-0.06	0.261	0.186	2	100%
century	21.07	21.03	-0.04	0.456	0.411	7	95%
Francis	19.26	19.23	-0.03	0.182	0.137	10	94%
picnic	17.94	17.91	-0.03	0.256	0.250	3	98%
roses	20.54	20.54	-0.00	0.208	0.203	4	98%

Table 22: Effect of mask erosion. Rows B and C isolate 4 px supervision-mask erosion using the same full evaluation masks. Row A shows the corresponding no-erosion result when evaluation masks are also eroded. Std: captured framing; 6×: widened framing.

	Configuration	Std PSNR	Std LPIPS	Zoom PSNR	Leak
A	no erosion, eval on eroded masks	19.86	0.170	18.25	0.221
B	no erosion, eval on full masks	**19.03**	**0.188**	**17.83**	0.206
C	4 px erosion, eval on full masks	18.37	0.201	17.63	**0.196**

14 MASK EROSION

Table 22 examines whether shrinking the supervision masks can suppress residual structure near object boundaries. Comparing Rows B and C, 4 px erosion reduces leakage from 0.206 to 0.196, but decreases standard PSNR from 19.03 to 18.37 dB and widened PSNR

from 17.83 to 17.63 dB. The main configuration therefore uses the uneroded masks, which better preserves target appearance for a small increase in leakage.

This trade-off is consistent with Gaussian rendering. A rendered silhouette arises from accumulated opacity rather than a sharp geometric boundary, so Gaussians whose centers lie near the 2D mask boundary can still contribute substantially to the target appearance. Erosion removes support from these boundary primitives first. In contrast, target-adjacent surfaces such as a ground slab or shelving surface can remain well inside the silhouette from multiple views and therefore survive even after the mask is eroded. Mask erosion consequently suppresses some boundary support without directly addressing the residual structure that is consistently aligned with the target.

15 TARGET-ADJACENT RADIANCE AND LEAKAGE

Figure 2 illustrates the complementary information provided by leakage and target photometrics. On *roses*, CleanGS achieves higher PSNR than GaussianCut on 44 of 46 frames, with an average advantage of 1.13 dB. The widened rendering also retains part of the illuminated tabletop immediately beneath the vase.

Because this region lies outside the bouquet-and-vase mask, it contributes to silhouette leakage. At the same time, the nearby radiance is visually coupled to the appearance of the transparent vase and its contact with the supporting surface; in the displayed frame, its removal coincides with a darker vase and a 1.61 dB lower widened PSNR for GaussianCut. Thus, off-mask opacity can capture both unwanted scene structure and target-adjacent appearance effects. Reporting leakage together with photometric fidelity distinguishes these different cleanup outcomes.

16 EFFECT OF ADDITIONAL GAUSSIANCUT SUPERVISION

Figure 3 shows the effect of increasing the supervision available to GaussianCut. With three selected masks, GaussianCut operates at the same filtering budget as CleanGS. Supplying all capture masks provides denser multiview evidence and removes additional residual structure in the examples shown. We therefore report the three-mask configuration as the matched-input comparison and the all-mask configuration, GaussianCut†, as an additional reference with substantially more supervision.

17 QUALITATIVE RESULTS FOR ALL SCENES

Figures 4 and 5 show one evaluation frame for every scene in the 18-scene benchmark at both captured framing and 6× widened field of view. Together, they provide the scene-level counterpart to the aggregate photometric and cleanup results, illustrating both target preservation and the residual structure exposed by reframing.

The widened reference is obtained by applying the same intrinsics transformation used for evaluation to the captured image. Because reference radiance is available only within the original captured field of view, the transformed ground truth occupies the corresponding central region of the widened image. Camera pose and output resolution remain fixed throughout.



Figure 2: Target-adjacent radiance and silhouette leakage. The *roses* scene at captured framing (top) and 6× widened framing (bottom). GaussianCut removes the illuminated tabletop beneath the bouquet and darkens the glass vase, while CleanGS preserves the vase, water, and nearby highlight. CleanGS is 1.61 dB better in widened PSNR on the displayed frame, although the retained tabletop region lies outside the binary target mask and therefore contributes to leakage.

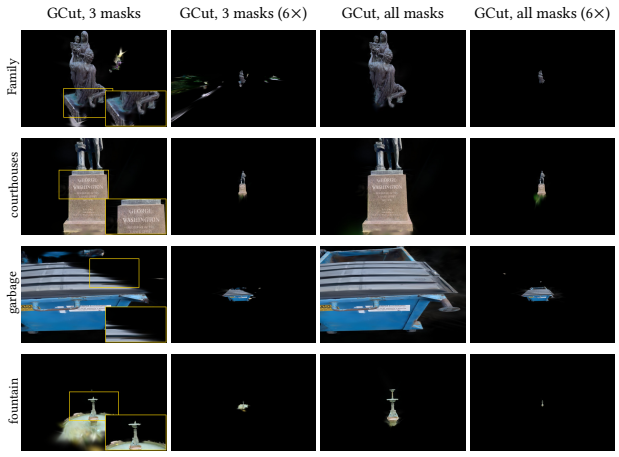


Figure 3: Effect of additional GaussianCut supervision. GaussianCut is shown with the same three selected masks used by CleanGS and with all available capture masks. Denser supervision removes additional residual structure and produces tighter silhouettes across these examples. The all-mask setting includes masks at scored cameras and corresponds to the daggered GaussianCut reference in the main paper.

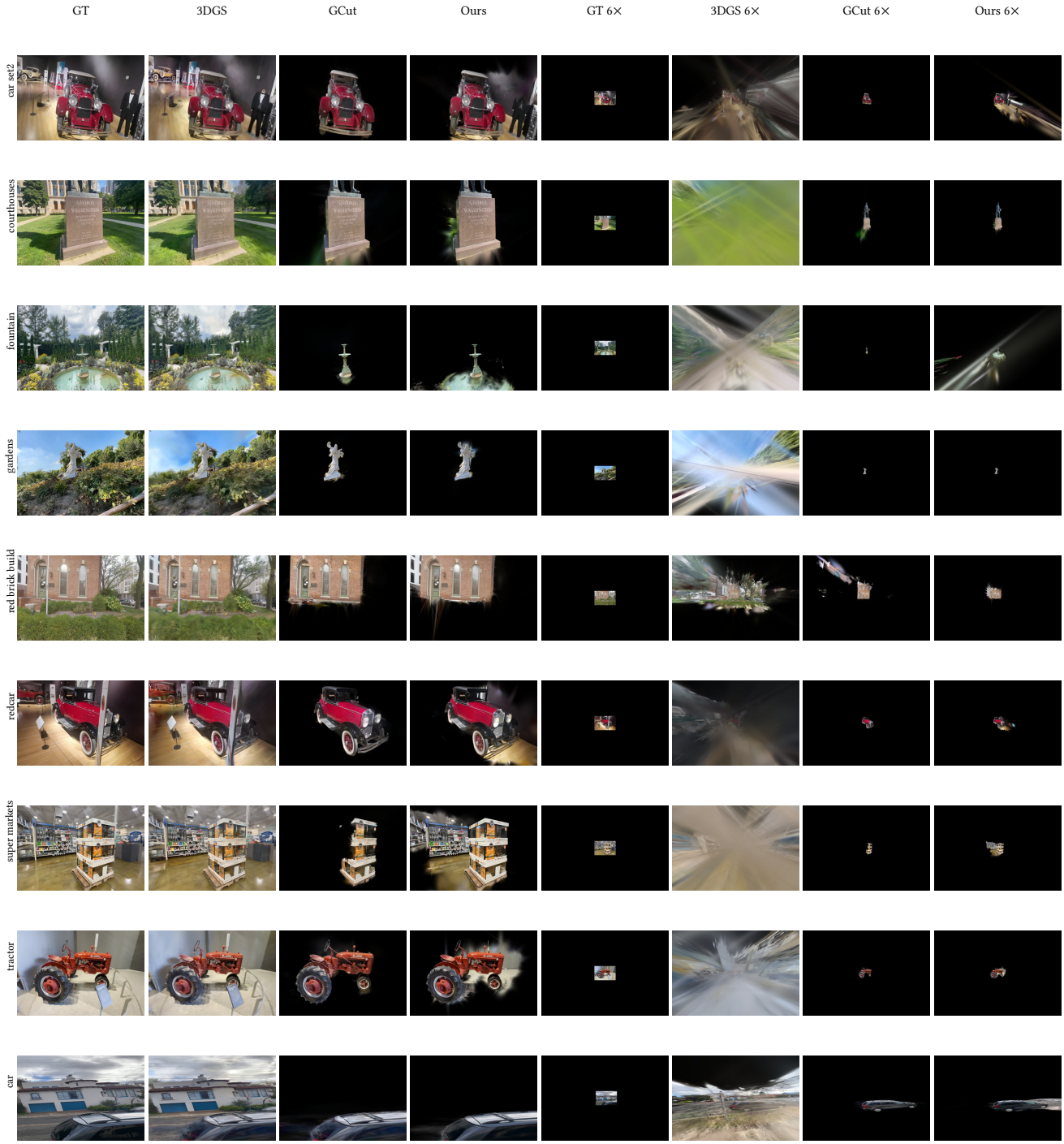


Figure 4: Qualitative results across DL3DV and NeRFBusters scenes. All eight DL3DV scenes and the NeRFBusters *car* scene are shown at captured and 6× widened framing. Widening preserves camera pose and output resolution while reducing focal length, exposing additional content retained in each exported asset. GaussianCut is shown using its all-mask GaussianCut† configuration.



Figure 5: Qualitative results across NeRFBusters and Tanks and Temples scenes. Six additional NeRFBusters scenes and all three Tanks and Temples scenes are shown at captured and 6× widened framing. The widened views preserve camera pose and output resolution while exposing Gaussian content retained outside the original framing. GaussianCut is shown using its all-mask GaussianCut† configuration.

REFERENCES

[1] Seokhun Choi, Hyeonseop Song, Jaechul Kim, Taehyeong Kim, and Hoseok Do. 2024. Click-Gaussian: Interactive Segmentation to Any 3D Gaussians. In *Computer Vision - ECCV 2024: 18th European Conference, Milan, Italy, September 29-October 4, 2024, Proceedings, Part III* (Milan, Italy). Springer-Verlag, Berlin, Heidelberg, 289–305. https://doi.org/10.1007/978-3-031-72646-0_17

[2] Richard Hartley and Andrew Zisserman. 2004. *Multiple View Geometry in Computer Vision* (2 ed.). Cambridge University Press.

[3] Wen Jiang, Boshu Lei, and Kostas Daniilidis. 2024. FisherRF: Active View Selection and Mapping with Radiance Fields Using Fisher Information. In *Computer Vision - ECCV 2024: 18th European Conference, Milan, Italy, September 29-October 4, 2024, Proceedings, Part XIII* (Milan, Italy). Springer-Verlag, Berlin, Heidelberg, 422–440. https://doi.org/10.1007/978-3-031-72624-8_24

[4] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. 2023. Segment Anything. arXiv:2304.02643 [cs.CV] https://arxiv.org/abs/2304.02643

[5] Juliette Marrie, Romain Menegaux, Michael Arbel, Diane Larlus, and Julien Mairal. 2025. LUDVIG: Learning-Free Uplifting of 2D Visual Features to Gaussian Splatting Scenes. In *IEEE/CVF International Conference on Computer Vision*. 7440–7450.

[6] Xuran Pan, Zihang Lai, Shiji Song, and Gao Huang. 2022. ActiveNeRF: Learning Where to See with Uncertainty Estimation. In *Computer Vision - ECCV 2022: 17th European Conference, Tel Aviv, Israel, October 23-27, 2022, Proceedings, Part XXXIII* (Tel Aviv, Israel). Springer-Verlag, Berlin, Heidelberg, 230–246. https://doi.org/10.1007/978-3-031-19827-4_14

[7] Johannes L. Schönberger and Jan-Michael Frahm. 2016. Structure-from-Motion Revisited. In *IEEE Conference on Computer Vision and Pattern Recognition*. 4104–4113.

[8] Johannes L. Schönberger, Enliang Zheng, Marc Pollefeys, and Jan-Michael Frahm. 2016. Pixelwise View Selection for Unstructured Multi-View Stereo. In *European Conference on Computer Vision*. 501–518.

[9] Hongyu Shen, Junfeng Ni, Yixin Chen, Weishuo Li, Mingtao Pei, and Siyuan Huang. 2025. Trace3D: Consistent Segmentation Lifting via Gaussian Instance Tracing. In *IEEE/CVF International Conference on Computer Vision*. 6656–6666.

[10] Wentao Sun, Quanyun Wu, Hanqing Xu, Kyle Gao, Zhengsen Xu, Yiping Chen, Dedong Zhang, Lingfei Ma, John S. Zelek, and Jonathan Li. 2025. SAGOnline: Segment Any Gaussians Online. https://doi.org/10.48550/arXiv.2508.08219 arXiv:2508.08219 [cs]